

COMPUTER NETWORKING PRACTICUM
APPLICATION LAYER PROTOCOL



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TI-21

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- **DIG Command**

```
debian@debian:~$ dig google.com

; <>> DiG 9.16.50-Debian <>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8575
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 4, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 0       IN      A      142.251.10.138
google.com.                 0       IN      A      142.251.10.113
google.com.                 0       IN      A      142.251.10.102
google.com.                 0       IN      A      142.251.10.101
google.com.                 0       IN      A      142.251.10.100
google.com.                 0       IN      A      142.251.10.139

;; AUTHORITY SECTION:
google.com.                 1255    IN      NS      ns1.google.com.
google.com.                 1255    IN      NS      ns3.google.com.
google.com.                 1255    IN      NS      ns4.google.com.
google.com.                 1255    IN      NS      ns2.google.com.

;; Query time: 4 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 10:44:34 WIB 2025
;; MSG SIZE rcvd: 196

debian@debian:~$ _
```

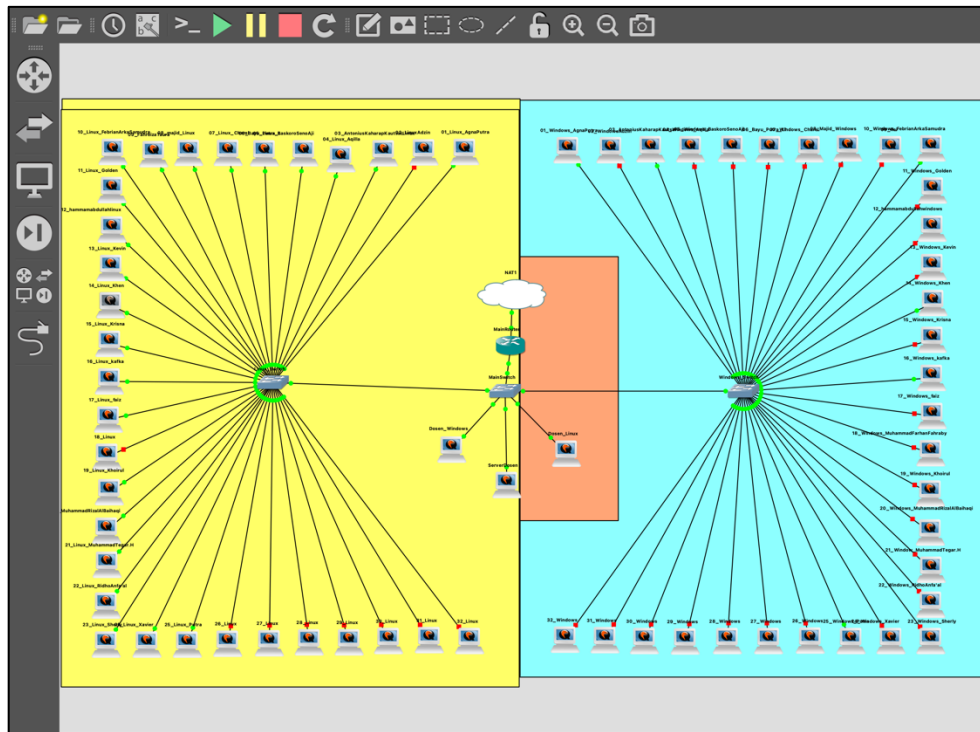
- **NSLOOKUP Command**

```
debian@debian:~$ nslookup google.com
Server:      10.10.10.1
Address:     10.10.10.1#53

Non-authoritative answer:
Name:   google.com
Address: 74.125.24.138
Name:   google.com
Address: 74.125.24.139
Name:   google.com
Address: 74.125.24.100
Name:   google.com
Address: 74.125.24.101
Name:   google.com
Address: 74.125.24.102
Name:   google.com
Address: 74.125.24.113
Name:   google.com
Address: 2404:6800:4003:c0f:64
Name:   google.com
Address: 2404:6800:4003:c0f:8b
Name:   google.com
Address: 2404:6800:4003:c0f:8a
Name:   google.com
Address: 2404:6800:4003:c0f:71

debian@debian:~$
```

PRACTICUM PREPARATION



14_Linux_Khen

vnc 192.168.60.24:5975

14_Windows_Khen

vnc 192.168.60.24:6007

PRACTICUM 1 (LINUX DIG & NSLOOKUP)

- **Check Internet Connection**

```
debian@debian:~$ ping google.com
PING google.com (74.125.24.100) 56(84) bytes of data.
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=1 ttl=53 time=27.2 ms
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=2 ttl=53 time=27.2 ms
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=3 ttl=53 time=27.1 ms
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=4 ttl=53 time=27.3 ms
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=5 ttl=53 time=27.1 ms
64 bytes from sf-in-f100.1e100.net (74.125.24.100): icmp_seq=6 ttl=53 time=27.1 ms
```

- **DIG Polinema**

```
debian@debian:~$ dig polinema.ac.id

; <<>> DiG 9.16.50-Debian <<>> polinema.ac.id
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 42544
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 0

;; QUESTION SECTION:
;polinema.ac.id.                IN      A

;; ANSWER SECTION:
polinema.ac.id.                310     IN      A      10.10.92.49

;; AUTHORITY SECTION:
polinema.ac.id.                431     IN      NS      ns3.polinema.ac.id.

;; Query time: 4 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 10:55:35 WIB 2025
;; MSG SIZE rcvd: 66
```

- **NSLOOKUP Polinema**

```
debian@debian:~$ nslookup polinema.ac.id
Server:          10.10.10.1
Address:         10.10.10.1#53

Non-authoritative answer:
Name:   polinema.ac.id
Address: 10.10.92.49
```

- **Find Ip Address and Server Name using DIG (Google)**

```

debian@debian:~$ dig google.com NS

; <<> DiG 9.16.50-Debian <<> google.com NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41669
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 4, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      NS

;; ANSWER SECTION:
google.com.                 452     IN      NS      ns4.google.com.
google.com.                 452     IN      NS      ns2.google.com.
google.com.                 452     IN      NS      ns1.google.com.
google.com.                 452     IN      NS      ns3.google.com.

;; AUTHORITY SECTION:
google.com.                 452     IN      NS      ns2.google.com.
google.com.                 452     IN      NS      ns1.google.com.
google.com.                 452     IN      NS      ns3.google.com.
google.com.                 452     IN      NS      ns4.google.com.

;; Query time: 0 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 10:57:56 WIB 2025
;; MSG SIZE rcvd: 156

```

- **Find Ip Address and Server Name using NSLOOKUP (Google)**

```

debian@debian:~$ nslookup -type=NS google.com
Server:          10.10.10.1
Address:         10.10.10.1#53

Non-authoritative answer:
google.com       nameserver = ns2.google.com.
google.com       nameserver = ns1.google.com.
google.com       nameserver = ns3.google.com.
google.com       nameserver = ns4.google.com.

Authoritative answers can be found from:
google.com       nameserver = ns1.google.com.
google.com       nameserver = ns3.google.com.
google.com       nameserver = ns4.google.com.
google.com       nameserver = ns2.google.com.

```

- **Find Ip Address and Server Name using DIG (Polinema)**

```

debian@debian:~$ dig polinema.ac.id NS

; <<> DiG 9.16.50-Debian <<> polinema.ac.id NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62437
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 0

;; QUESTION SECTION:
;polinema.ac.id.          IN      NS

;; ANSWER SECTION:
polinema.ac.id.           4113    IN      NS      ns3.polinema.ac.id.

;; AUTHORITY SECTION:
polinema.ac.id.           4113    IN      NS      ns3.polinema.ac.id.

;; Query time: 0 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 12:11:16 WIB 2025
;; MSG SIZE rcvd: 64

```

- Find Ip Address and Mail Server Name using DIG (Google)

```

debian@debian:~$ dig google.com MX

; <<> DiG 9.16.50-Debian <<> google.com MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12812
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      MX

;; ANSWER SECTION:
google.com.                108     IN      MX      10 smtp.google.com.

;; Query time: 0 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 12:13:53 WIB 2025
;; MSG SIZE rcvd: 49

```

- Find Ip Address and Mail Server Name using NSLOOKUP (Google)

```

debian@debian:~$ nslookup -type=MX google.com
Server:          10.10.10.1
Address:         10.10.10.1#53

Non-authoritative answer:
google.com       mail exchanger = 10 smtp.google.com.

Authoritative answers can be found from:

```

- Find Ip Address and Mail Server Name using DIG (Polinema)

```

debian@debian:~$ dig polinema.ac.id MX

; <<> DiG 9.16.50-Debian <<> polinema.ac.id MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 32136
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 1, ADDITIONAL: 0

;; QUESTION SECTION:
;polinema.ac.id.          IN      MX

;; ANSWER SECTION:
polinema.ac.id.          4462    IN      MX      5 alt1.aspmx.l.google.com.
polinema.ac.id.          4462    IN      MX      1 aspmx.l.google.com.
polinema.ac.id.          4462    IN      MX      10 alt4.aspmx.l.google.com.
polinema.ac.id.          4462    IN      MX      10 alt3.aspmx.l.google.com.
polinema.ac.id.          4462    IN      MX      5 alt2.aspmx.l.google.com.

;; AUTHORITY SECTION:
polinema.ac.id.          3864    IN      NS      ns3.polinema.ac.id.

;; Query time: 0 msec
;; SERVER: 10.10.10.1#53(10.10.10.1)
;; WHEN: Thu Feb 27 12:15:24 WIB 2025
;; MSG SIZE rcvd: 168

```

- Find Ip Address and Mail Server Name using NSLOOKUP (Polinema)

```
debian@debian:~$ nslookup -type=MX polinema.ac.id
Server:      10.10.10.1
Address:     10.10.10.1#53

Non-authoritative answer:
polinema.ac.id mail exchanger = 1 aspmx.l.google.com.
polinema.ac.id mail exchanger = 10 alt4.aspmx.l.google.com.
polinema.ac.id mail exchanger = 10 alt3.aspmx.l.google.com.
polinema.ac.id mail exchanger = 5 alt2.aspmx.l.google.com.
polinema.ac.id mail exchanger = 5 alt1.aspmx.l.google.com.

Authoritative answers can be found from:
polinema.ac.id nameserver = ns3.polinema.ac.id.
```

PRACTICUM 2 (LINUX SSH)

- **Connect to Class SSH**

```
debian@debian:~$ ssh ti2i@10.10.10.10
The authenticity of host '10.10.10.10 (10.10.10.10)' can't be established.
ECDSA key fingerprint is SHA256:dyt3wAxrjU6U4nqrbz5YMPFLuPiDqY/Awfu4YNn1llw.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.10' (ECDSA) to the list of known hosts.
ti2i@10.10.10.10's password:
Linux debian 5.10.0-8-amd64 #1 SMP Debian 5.10.46-4 (2021-08-03) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Feb 27 12:13:47 2025 from 10.10.10.14
ti2i@debian:~$ _
```

- **Creating Directory**

```
ti2i@debian:~$ mkdir KhenCahyo_244107023002
ti2i@debian:~$ ls
2341720205_Charel      BayuPutraL_2341720240    Kevin_2341720233
Ade_10101010          Charel_2341720205       KhenCahyo_244107023002
```

- **Moving .txt File**

```
ti2i@debian:~$ cp /home/ti2i/KhenCahyo_244107023022.txt ./KhenCahyo_244107023002
ti2i@debian:~$ cd KhenCahyo_244107023002
ti2i@debian:~/KhenCahyo_244107023002$ ls
KhenCahyo_244107023022.txt
ti2i@debian:~/KhenCahyo_244107023002$
```

- **Edit File Content**

```
GNU nano 5.4                                KhenCahyo_244107023022.txt
Name: Khen Muhammad Cahyo
NIM: 244107077023022
Department: Information Technology
```

- **Logout From SSH**

```
ti2i@debian:~/KhenCahyo_244107023002$ exit
logout
Connection to 10.10.10.10 closed.
debian@debian:~$
```


PRACTICUM 2 (LINUX FTP)

- **Creating New File**

```
GNU nano 5.4 KhenCahyo_FTP.txt
Name: Khen Muhammad Cahyo
NIM: 244107023002
Department: Information Technology
```

- **Installing FTP**

```
debian@debian:~$ sudo apt install ftp
[sudo] password for debian:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ftp
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 59.9 kB of archives.
After this operation, 140 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian bullseye/main amd64 ftp amd64 0.17-34.1.1 [59.9 kB]
Fetched 59.9 kB in 0s (596 kB/s)
Selecting previously unselected package ftp.
(Reading database ... 28243 files and directories currently installed.)
Preparing to unpack .../ftp_0.17-34.1.1_amd64.deb ...
Unpacking ftp (0.17-34.1.1) ...
Setting up ftp (0.17-34.1.1) ...
update-alternatives: using /usr/bin/netkit-ftp to provide /usr/bin/ftp (ftp) in auto mode
Processing triggers for man-db (2.9.4-2) ...
debian@debian:~$ _
```

- **Getting File From FTP**

```
ftp> get KhenCahyo_244107023022.txt
local: KhenCahyo_244107023022.txt remote: KhenCahyo_244107023022.txt
200 PORT command successful.
150 Opening BINARY mode data connection for 'KhenCahyo_244107023022.txt' (82 bytes).
226 Transfer complete.
82 bytes received in 0.00 secs (567.9300 kB/s)
ftp> bye
221 Goodbye.
debian@debian:~$ _
```

- **Putting File From FTP**

```
ftp> put KhenCahyo_244107023022.txt
local: KhenCahyo_244107023022.txt remote: KhenCahyo_244107023022.txt
200 PORT command successful.
150 Opening BINARY mode data connection for 'KhenCahyo_244107023022.txt'.
226 Transfer complete.
82 bytes sent in 0.00 secs (964.7967 kB/s)
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for '/bin/ls'.
total 4
-rw-r--r-- 1 ti2i ti2i 82 Feb 27 13:37 KhenCahyo_244107023022.txt
226 Transfer complete.
ftp>
```

PRACTICUM 1 (WINDOWS NSLOOKUP)

- Check Connection

```
C:\Documents and Settings\XP>ping polinema.ac.id

Pinging polinema.ac.id [10.10.92.49] with 32 bytes of data:

Reply from 10.10.92.49: bytes=32 time=1ms TTL=60
Reply from 10.10.92.49: bytes=32 time=3ms TTL=60
Reply from 10.10.92.49: bytes=32 time=1ms TTL=60
Reply from 10.10.92.49: bytes=32 time=1ms TTL=60
```

- NSLOOKUP Polinema

```
C:\Documents and Settings\XP>nslookup polinema.ac.id
*** Can't find server name for address 10.10.10.1: Non-existent domain
*** Default servers are not available
Server: Unknown
Address: 10.10.10.1

Non-authoritative answer:
Name:   polinema.ac.id
Address: 10.10.92.49

C:\Documents and Settings\XP>_
```

- Find Ip Address and Server Name using Polinema

```
C:\Documents and Settings\XP>nslookup -type=NS polinema.ac.id
*** Can't find server name for address 10.10.10.1: Non-existent domain
*** Default servers are not available
Server: Unknown
Address: 10.10.10.1

Non-authoritative answer:
polinema.ac.id nameserver = ns3.polinema.ac.id
polinema.ac.id nameserver = ns3.polinema.ac.id

C:\Documents and Settings\XP>
```

- Find Ip Address and Mail Server Name Polinema

```
C:\Documents and Settings\XP>nslookup -type=MX polinema.ac.id
*** Can't find server name for address 10.10.10.1: Non-existent domain
*** Default servers are not available
Server: Unknown
Address: 10.10.10.1

Non-authoritative answer:
polinema.ac.id MX preference = 5, mail exchanger = alt1.aspmx.l.google.com
polinema.ac.id MX preference = 5, mail exchanger = alt2.aspmx.l.google.com
polinema.ac.id MX preference = 10, mail exchanger = alt3.aspmx.l.google.com
polinema.ac.id MX preference = 10, mail exchanger = alt4.aspmx.l.google.com
polinema.ac.id MX preference = 1, mail exchanger = aspmx.l.google.com

C:\Documents and Settings\XP>
```